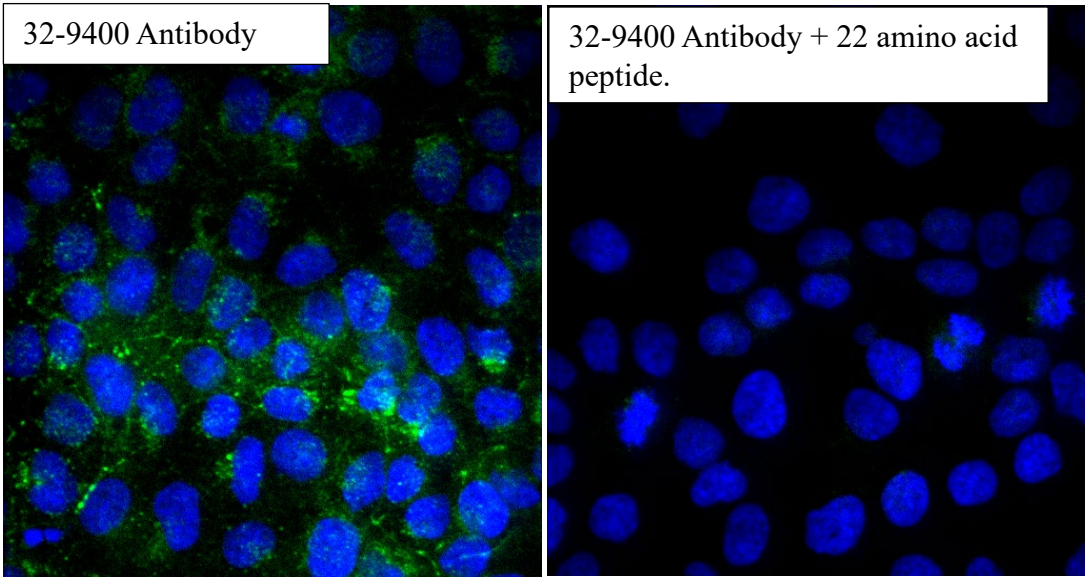


Claudin-4 Localization in Epithelial Ovarian Cancer

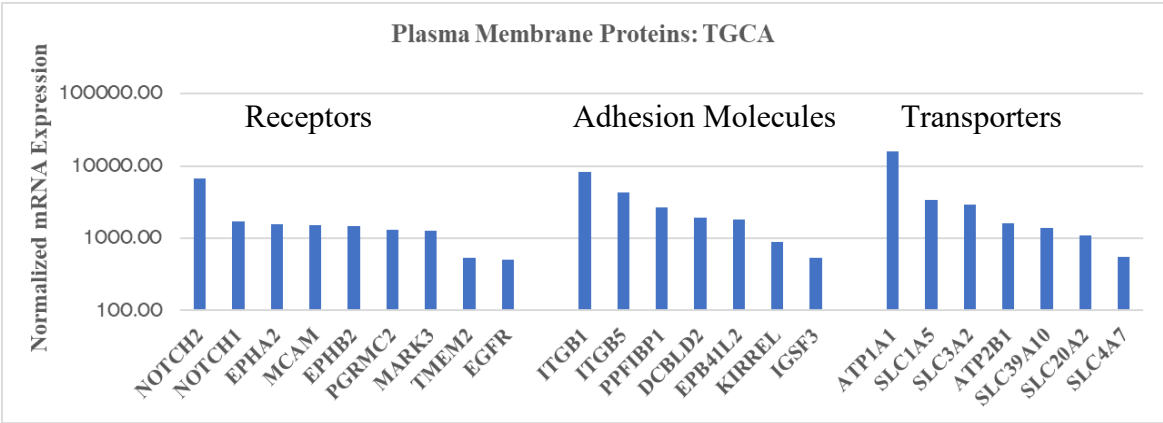
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Supplementary Material

Supplemental Figure 1. Evaluation of claudin-4 antibody. Thermo-Fisher Monoclonal against 22 amino acid C-terminus of claudin-4. Staining is completely abrogated in the presence of a 22 amino acid peptide mimic of the last 22 amino acids of human claudin-4.



Supplementary Figure 2. mRNA levels from the TCGA data for ovarian cancer for the plasma membrane proteins identified in the claudin-4 BioID experiment.



Supplementary Table 1. Average data from claudin-4 BioID experiments

Symbol	Intensity, control	Intensity C CLDN4	Average Ratio	Log2 Ratio	Average Difference	Number of experiments
ADD3	2.00	14.00	8.60	4.94	12	3
AHNAK	1688.00	3529.50	2.04	1.15	1842	4
AKAP12	58.75	224.50	3.39	2.09	166	4
AP1B1	1.00	30.25	30.25	8.08	29	4
ATP1A1	8.75	42.25	9.20	3.16	34	4
ATP2B1	1.00	17.00	17.00	5.77	16	3
BAIAP2	2.50	36.00	28.07	7.11	34	4
CPD	1.00	21.50	21.50	6.19	21	4
CTNND1	15.50	128.50	38.90	5.50	113	4
CXADR	1.00	23.75	23.75	8.29	23	4
DCBLD2	1.00	15.00	15.00	3.89	14	3
DLG1	6.00	22.50	6.79	3.03	17	3
DSC2	1.00	47.33	47.33	5.42	46	3
DSG2	6.00	67.75	14.18	6.19	62	3
EGFR	13.75	72.50	20.29	5.43	59	4
EHD1	7.00	18.75	9.47	5.02	12	4
EPB41L2	19.75	77.00	14.93	3.16	57	4
EPHA2	5.50	94.75	46.43	19.11	89	4
EPHB2	1.00	26.75	26.75	9.49	26	4
ERBIN	36.75	69.25	8.00	2.60	33	3
FERMT2	8.00	48.75	10.59	4.56	41	4
GMAP210	8.25	48.00	9.81	3.96	40	4
GOLGB1	22.50	83.50	3.47	2.12	61	4
IGF2R	8.00	29.00	8.97	6.03	21	3
IGSF3	1.00	22.50	22.50	6.04	22	4
ITGB1	2.00	80.50	35.30	12.10	79	4
ITGB5	1.00	35.50	35.50	7.07	35	4
JUP	12.75	30.00	4.73	3.89	17	3
KIRREL1	1.00	33.75	33.75	11.63	33	4
LEMD3	2.25	26.00	12.65	8.96	24	3
LNPEP	1.00	28.00	28.00	6.45	27	4
LSR	1.00	15.00	15.00	4.58	14	3
MARCKS	1.25	16.00	14.25	6.75	15	4
MARK3	1.00	19.75	19.75	7.20	19	4
MCAM	1.75	72.25	31.94	13.00	71	4
NOTCH1	1.00	15.25	15.25	6.02	14	3
NOTCH2	6.00	64.75	26.18	12.66	59	4
OCLN	1.00	15.75	15.75	4.93	15	3
PGRMC2	12.25	48.75	4.03	2.63	37	3
PHACTR4	11.50	33.00	8.57	2.95	22	3

PPFIBP1	12.00	49.50	8.11	3.62	38	3
RAB7A	3.00	40.25	22.04	9.03	37	4
SLC1A5	2.25	34.50	23.67	8.53	32	4
SLC20A2	1.00	12.50	12.50	5.72	12	3
SLC7A5	1.67	12.67	9.78	3.11	11	4
SLC39A10	1.00	23.75	11.08	7.14	12	3
SLC3A2	5.25	99.25	35.75	19.34	35	4
SLC4A7	4.00	58.00	28.26	6.44	81	3
SNAP23	4.00	33.75	33.45	4.12	30	4
SPTBN2	1.00	75.25	75.25	9.14	74	4
TMEM2	1.00	27.75	27.75	6.28	27	4
TRIP11	8.25	48.00	9.81	3.96	40	4
YES1	3.00	21.00	9.60	4.53	18	3
YKT6	3.25	60.50	19.65	7.72	57	4

Supplementary Table 2A. BioID identified proteins, nucleus, ER and Golgi		
Symbol	Protein name	Function
Nucleus		
LEMD3	LEM domain–containing protein 3	Connected to the actin-microfilament network through the LINC (nesprin-sun) complex; down regulates TGF-beta/SMAD signaling (Chambers et al., 2018).
ER and Golgi		
GMAP210: TRIP11	Golgin GMAP 210	Tethers vesicles to the Golgi via ARF:GTP at the ER-Golgi intersection and in the Golgi complex (Infante et al., 1999).
GOLGB1	Giantin; Golgin subfamily B member 1	Transmembrane protein localized in the medial Golgi with large cytoplasmic coiled-coil regions (Linstedt and Hauri, 1993).
AP1B1	AP-1 complex subunit beta-1	Clathrin-associated adaptor protein in the trans-Golgi; plays a role in sorting endosomes (Anitei et al., 2010).
IGF2R	Insulin like growth factor receptor 2	Transmembrane receptor: binds mannose-6-phosphate tagged proteins in the trans-Golgi (Brown et al., 2008).
CPD	carboxypeptidase D	Trans-Golgi network protein; releases c-terminal arginine and lysine from polypeptides (Thomas et al., 2015).
YES1	Yamaguchi sarcoma viral oncogene homolog	Non-membrane spanning protein with tyrosine kinase activity associated with cytoplasm and Golgi (Zhou et al., 2020).

Supplementary Table 2B. BioID identified proteins, endosomal proteins		
LNPEP	Also called IRAP. Leucyl/cysteinyl aminopeptidase	The cytosolic portion of the membrane protein LNPEP is an endosomal trafficking adaptor (Descamps et al., 2020).
EHD1	EH domain-containing protein 1	Transmembrane ATPase that regulates several steps of endosomal recycling (Dhawan et al., 2020, Deo et al., 2018).
RAB7a	Ras-related protein Rab-7a	Rab-7a is a key regulator of endosome-lysosome vesicular trafficking (Guerra and Bucci, 2019, Modica et al., 2017).
SNAP23	Synaptosome associated protein 23	SNAP23 is a SNARE regulator of transport vesicle docking and fusion at the plasma membrane (Sun et al., 2016)
YKT6	Synaptobrevin homolog	This membrane associated protein regulates vesicle trafficking throughout the cell (Gordon et al., 2017).

Supplementary Table 2C. BioID identified proteins, proteins associated with cell-cell junctions		
LSR	Lipolysis-stimulated lipoprotein receptor	The membrane protein LSR ferries lipid molecules into cells and localizes tricellulin at tricellular junctions (Masuda et al., 2011).
OCLN	Occludin	OCLN is a tetraspanin membrane protein often involved in tight junctions in epithelia (Fredriksson et al., 2015).
CTNND1	Delta 1 catenin; p120-catenin	The plasma membrane protein P120-catenin interacts with E-cadherin stabilizing the cell-cell adherens junction (Ishiyama et al., 2010).
DLG1	DLG1; discs large scaffold protein 1	DLG1 is a scaffolding MAGUK (membrane associated guanylate kinase) protein (Marziali et al., 2019).
CXADR	Coxsackievirus and adenovirus receptor (CAR)	CAR is a type I transmembrane glycoprotein that serves as a junction-adhesion molecule (Nilchian et al., 2019).
DSC2, DSG2	Desmocollin2 Desmoglein2	Two desmosomal membrane proteins that interact in the space between cells (Green et al., 2019).
JUP	Junction plakoglobin	JUP is sub-membranous and associated with desmosomes and adherens junctions (Aktary et al., 2017, Weiland et al., 2020).

Table 2D. BioID identified proteins, cytoskeletal and scaffolding proteins.		
SPTBN2	Beta III spectrin protein	Interaction with actin stabilizes transporter EAAT4 at the plasma membrane (Machnicka et al., 2019).
ADD3	Adducin 3	Promotes assembly of the spectrin-actin sub-membrane network (Kiang and Leung, 2018).
MARCKS	Myristoylated alanine-rich C-kinase substrate	A target of Protein kinase C that modulates cytoskeletal actin dynamics and vesicular trafficking (Iyer et al., 2021).
BAIAP2; IRSp53	Brain-specific angiogenesis inhibitor 1-associated protein 2; insulin receptor substrate p53	BAIAP2 links membrane bound G-proteins to cytoplasmic effector proteins such as Rho (Liu et al., 2010, Funato et al., 2004).
PHACTR4	phosphatase and actin regulator 4	Binds protein phosphatase 1 and actin, maintaining the cellular pool of actin (Huet et al., 2013).
AKAP12	A-kinase anchoring protein 1, gravin, AKAP250	AKAPS provide an organizing center for G-protein coupled receptors (Malbon et al., 2004).
ERBIN	ERBB2 Interacting Protein; ERBB2IP	A scribble homolog that stabilizes the unphosphorylated form of ERBB2 (Santoni et al., 2020).
FERMT2	Kindlin-2	A pleckstrin homology protein required for focal adhesion assembly (Kadry et al., 2018).

Supplementary Table 2E. BioID identified proteins, plasma membrane receptors		
NOTCH2 NOTCH1	Notch receptor 2, Notch receptor 1	Activated by DLL and JAG on mesenchymal cells (Jia et al., 2019, Alniaimi et al., 2015, Wang et al., 2010).
EPHA2	EPH receptor A2	Receptor tyrosine kinase activated by PI3K. Regulates integrin expression (Takahashi et al., 2018).
EPHB2	EPH receptor B2	Modifies the actin cytoskeleton and alters the activities of integrins (Liu et al., 2022).
EGFR	Epidermal growth factor receptor	Transmembrane glycoprotein binding EGF; significant cross talk with CDH1 (Ramírez Moreno and Bulgakova, 2021).
MARK3	Microtubule affinity regulating kinase 3	Interacts with ARHGEF2 to promote organized cellular architecture (Sandí et al., 2017, Machino et al., 2022).
MCAM	Melanoma cell adhesion molecule; CD146	Transmembrane protein functions as a cell receptor for many signaling molecules (Wang et al., 2020).
PGRMC2	Progesterone Receptor Membrane Component Type 2	PGRMC2 has cytochrome P450 (CYP) domains that bind heme (Ryu et al., 2017). Regulated by PPAR- γ in ovarian cancer cells (Shen et al., 2020).
TMEM2; CEMIP2	Cell migration inducing hyaluronidase 2	Cleaves extracellular hyaluronic acid before internalization into lysosomes (Sinha et al., 2014, Schinzel et al., 2019).

Supplementary Table 2F. BioID identified proteins, plasma membrane adhesion molecules		
ITGB1	Integrin beta1	ITGB1 is a membrane-localized collagen receptor associated with high recurrence free survival in the TGCA (Zhu et al., 2020).
ITGB5	Integrin beta 5	ITGB5 appears correlated with FAK in anchorage independent ovarian cancer cell growth (Tancioni et al., 2014). Interactions between claudins and integrins may regulate cell-matrix interactions and cell death (Ding et al., 2012)
PPFIBP1	Liprin-beta-1	Interacts with s100A4, a calcium binding protein. High expression correlates with poorer overall survival in ovarian cancer (Behera et al., 2020).
DCBLD2	Discoidin, CUB and LCCL domain-containing protein 2	DCBLD2 interacts with hormone and growth factor receptors (Schmoker et al., 2019).
EPB41L2	Erythrocyte Membrane Protein Band 4.1 Like 2	A common variant of the erythrocyte protein Band 4; predicts poor survival in HGSOE (Danan-Gotthold et al., 2015).
IGSF3	Immunoglobulin superfamily member 3	IGSF3 can activate NF- κ B to promote tumor progression (Sheng et al., 2020).
KIRREL1	Kirre like nephrin family adhesion molecule 1	KIRREL1 regulates cytoskeletal organization; Expression levels are related to proliferation, metastasis, and poor cancer prognosis (Chen et al., 2020, Garg et al., 2007).

TABLE 2G. BioID identified proteins, plasma membrane transporters		
ATP1A1	NA⁺/K⁺-transporting ATPase subunit alpha-1	Maintains electrochemical gradients of Na ⁺ and K ⁺ across the plasma membrane (Huang et al., 2020).
ATP2B1	Plasma membrane calcium-transporting ATPase 1; PMCA1	Plays a critical role in intracellular calcium homeostasis (Stafford et al., 2017).
SLC1A5	ASCT2, Neutral amino acid transporter B	A glutamate transporter important in many metabolic processes (Lopes et al., 2021, Guo et al., 2018).
SLC3A2	4F2hc, partner of SLC7A7	Upregulation of SLC3A2 decreased the growth of ovarian tumors (Yan et al., 2021).
SLC7A5	LAT1, Essential amino acid transporter	Highly expressed in epithelial ovarian cancer and correlates with a poor prognosis (Kaira et al., 2015).
SLC20A2	Sodium-dependent phosphate transporter 2, PIT2	Not differentially expressed in ovarian cancer (Böttger and Pedersen, 2011).
SLC39A10	Zinc transporter, ZIP10	High expression of ZIP10 is associated with slightly increased survival (Human Protein Atlas) (Bafaro et al., 2017).
SLC4A7	Sodium bicarbonate transporter 3, NBCN1, NBC3	An exchanger for Na ⁺ and HCO ³ important in maintenance of intracellular pH (Parker and Boron, 2013).

Supplementary Table 3 Primary Antibodies		
Antibody	Source	Catalog Number
claudin 4	Invitrogen	32-9400
phospho-claudin 4 (Tyr208)	Invitrogen	PA5-105982
phospho-FAK (Y397)	BD Transduction Laboratories	BD611722
GM130	Cell Signaling	12480
GMAP210	Abcam	ab251664
Giantin	Abcam	ab37266
E cadherin	Cell Signaling	3195
delta catenin	ThermoFisher	33-9600
EGFR	Cell Signaling	4267
integrin beta 1	R and D Systems	MAB17781
RAB7	Invitrogen	PA5-52369
YKT6	Abcam	ab236583
NOTCH2	Cell Signaling	5732
EPHA2	Cell Signaling	6997
EPHB2	Cell Signaling	83029
DLG1	Invitrogen	PA1-741
LSR	Novus Biologicals	NBP1-89631